

Usage of Optical Correlator in Video Surveillance System for Abandoned Luggage

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Abstract—This article is focused on new methods of image processing used in modern video surveillance systems. The aim of this article is usage of the latest knowledge in the field of digital signal processing (image and video) and optical correlation in the design and experimental verification of video surveillance system that should detect lost or abandoned luggage.

Index Terms—Image processing, optical correlator, video surveillance system

I. INTRODUCTION

Nowadays, the evolution of security is even faster with help of science and new technology. A significant development in computer technology in recent decades also reflected in the development of safety systems. A simple camera system linked to multiple displays allows security officer to monitor the situation in the monitored area in real time is now replaced by a much more comprehensive solution. The analog signal is replaced by digital one and it is generally treated by control server. The software suite enables intelligent monitoring of the area, face recognition of unauthorized persons or detection of illegal activity. Algorithms used in such applications generally use methods and techniques of digital signal processing and optical correlation [1, 2].

Our goal was to create a test system for detecting abandoned luggage, which would employ above mentioned technologies. The system can detect objects in the input image (person, luggage) and track their movement (by monitoring their coordinates).

II. DESIGN OF VIDEO SURVEILLANCE SYSTEM FOR ABANDONED LUGGAGE WITH CAMBRIDGE OPTICAL CORRELATOR

One of the possible methods for object recognition in video surveillance systems used in security applications can also be optical correlation. The aim of these applications is mainly to support and facilitate the work of security personnel in various buildings such as airports, subway, banks, casinos etc. Their task is to detect presence and movement of people and things, recognize and classify patterns in video footage. Subsequent analysis of acquired data can alert staff to potential threats

(Threat Detection), the presence of unauthorized persons (Intrusion Detection) etc.

This designed and tested system consists of two main parts, hardware and software part. The hardware portion consists of following blocks:

- camera system,
- optical correlator,
- server (Fig. 1).

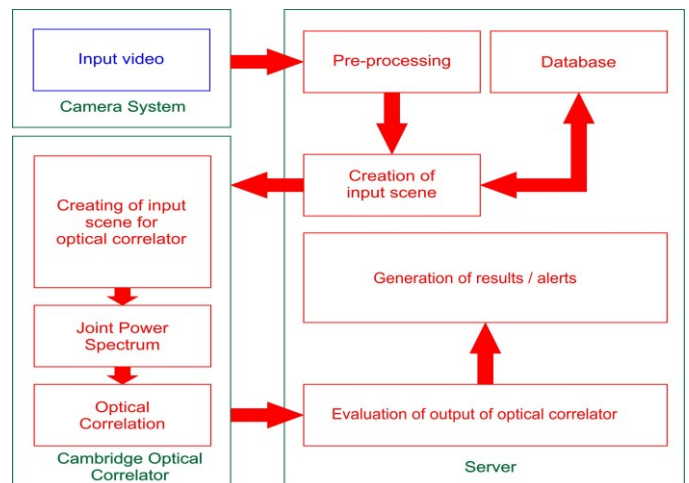


Fig. 1. Block diagram of video surveillance system.

The input video is captured by the camera system, consisting of one or more IP cameras connected via network infrastructure to other elements of the system. Server provides computing resources for the software part of this system. It consists (depending on the purpose of the system) of the following blocks:

- pre-processing of the input video,
- database (contains reference samples, resp. models used for correlation),
- preparation of the input image before the optical correlation and communication with the optical correlator (using the API - Application Programming Interface),
- evaluation of the output of the optical correlator,

- results generating, resp. alerts (depending on what the video control system is used).

Cambridge optical correlator belongs to a group of JTC also referred to as 1/f Phase-Only Joint Transform Correlator. The expression 1/f indicates only a simple Optical Fourier Transform in both stages of the process JTC. This means using the same optics twice in the process of correlation.

Block pre-processing includes various methods of digital image processing (gamma correction, color filtration, edge detection, etc.) to ensure better quality of the input of the optical correlator. The result of correlation is obtained values of similarity, respectively match of the image captured by the camera system and the reference image from the database, as well as the coordinates of objects in the entrance plane [3, 4, 5, 6].

III. EXPERIMENTS AND RESULTS

Basis of proposed video surveillance system is evaluation of the distance of objects detected in the monitored area. The sequence of input images is pre-processed by using applications designed and programmed in C#. Then they are analyzed by Cambridge optical correlator. Based on the coordinates of these objects we can determine whether the subject tries to leave the luggage.

A. Obtaining of experimental images

To simplify the experiment, input video was temporarily replaced by a sequence of images. The experiment was carried out with four sets of frames on which are generally three types of luggage and two people. Each set contains five frames and each frame consists of a luggage, which is still in the same position and the man who is gradually moving away from it (Fig. 2).

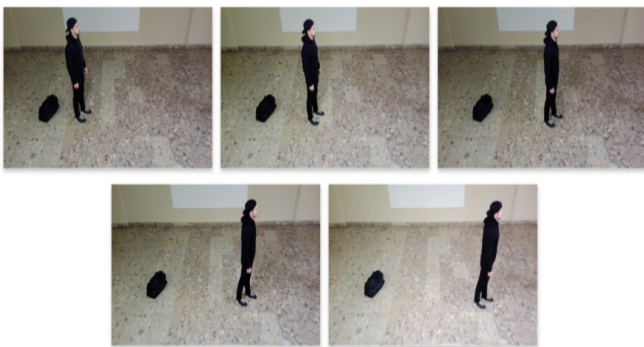


Fig. 2. Example of sequence of input images.

First, each image is pre-processed (Fig. 3) and it consists of:

- Median filter (Fig. 3a) - noise reduction of the input images,
- gamma correction (Fig. 3a) - adjustment of values of color components,
- color filtration (Fig. 3b) - detection of objects based on certain colors, in this case mainly black,
- converting the image into the grayscale (Fig. 3c),

- edge detection using Sobel operator (Fig. 3d),
- blob filtration (blob in this case is a white object on a black background) under the minimum size (Fig. 3e),
- filter to fill the remaining objects in white (Fig. 3f).

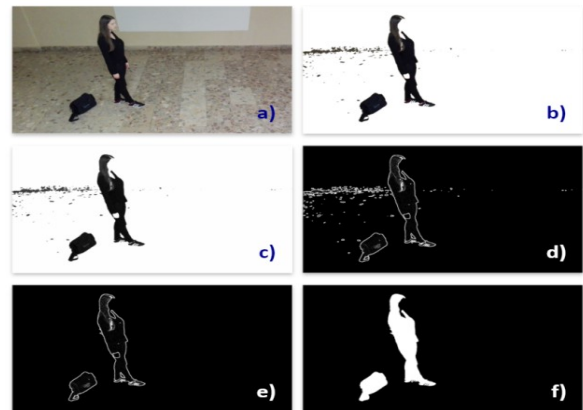


Fig. 3. Pre-processing of input frame.

The result of the pre-processing by using several filters (gamma correction, color filter, edge detection, etc.), and functions is the image that is suitable for further processing by the optical correlator. All images are stored in the destination directory from which they are subsequently loaded into Fourier Optics Experimenter. The sequence of images after pre-processing is shown in Fig. 4.

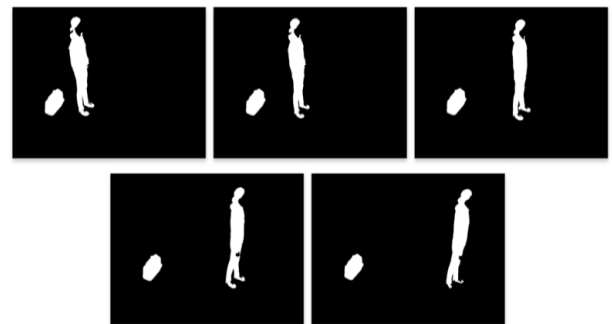


Fig. 4. Sequence of input frames after pre-processing.

Optical correlation is then performed between the first frame and the first frame and the first frame and any other frames. In terms of design, the most important data are the changes of coordinates of objects between frames. Based on these data the system should detect abandoned baggage, resp. a person who tries to abandon this baggage.

B. Evaluation of experiments

Cambridge optical correlator has an input and reference image aligned and displayed together on the Spatial Light Modulator. Subsequently, these images are transformed by Fourier transformation and a non-linear camera then captures the intensity distribution of transform to produce Joint Power Spectrum (JPS). JPS is then binary or threshold processed and this processed image enters to transform process as input image of second FT. Output of this transform is the correlation plane that includes correlation peaks per match.

Fig. 5 shows the change of the position of the correlation peaks in the output plane of the optical correlator based on the change in position of an object (white squares) in the entry level.

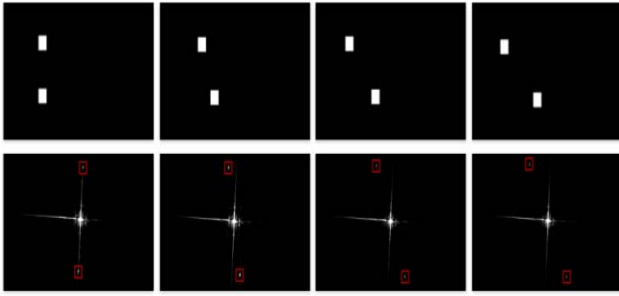


Fig. 5. Example of sequence of input images.

Obtained coordinates of correlation peaks are shown in TABLE I. The upper left corner of the output plane has coordinates $[0, 0]$. Fig. 5 and TABLE I. show that if you increase the distance of objects in the input plane in the direction of the X-axis, the distance of correlation peak at the output increases in the same direction (coordinate difference Δx). The same principle would apply also when to another position in the opposite direction, respectively the Y-axis direction.

TABLE I. COORDINATES OF CORRELATION PEAKS

Test	x_1	y_1	x_2	y_2	Δx
1.	257	213	243	384	14
2.	235	210	265	386	30
3.	213	207	288	389	76
4.	201	206	299	390	98

The process of optical correlation (Fig. 6) consists of the following steps:

- loading of images into the input plane (Fig. 6a),
- selection of the ROI to obtain spectrum JPS (Fig. 6b),
- setting of parameters of CMOS camera
- selection of the ROI of correlation peaks (Fig. 6d),
- obtaining the coordinates of the correlation peaks.

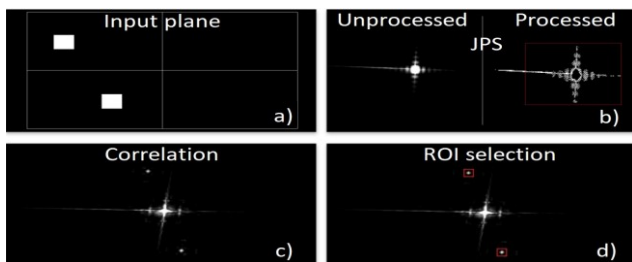


Fig. 6. Process of optical correlation.

The principle of obtaining the coordinates of the object is used in the design of video surveillance systems for abandoned luggage. Based on these coordinates it is also possible to evaluate whether the objects are closer to or any from each

other. So, the system is able to warn whether a person is attempting to leave luggage in the monitored area.

Fig. 7 shows the change of the position of the correlation peaks in the output of the optical correlator (ROIs are highlighted in red). The figures show that if you change the position of objects in the input plane, position of the correlation peaks in the output plane of correlator is also changed. The coordinates of these peaks are obtained by analyzer. Their comparison enables tracking the movement of objects in the monitored area, and also enables measuring the distance between them. In the case that the monitored person is leaving from luggage in the opposite direction, position of correlation peak was mirror-inverted.

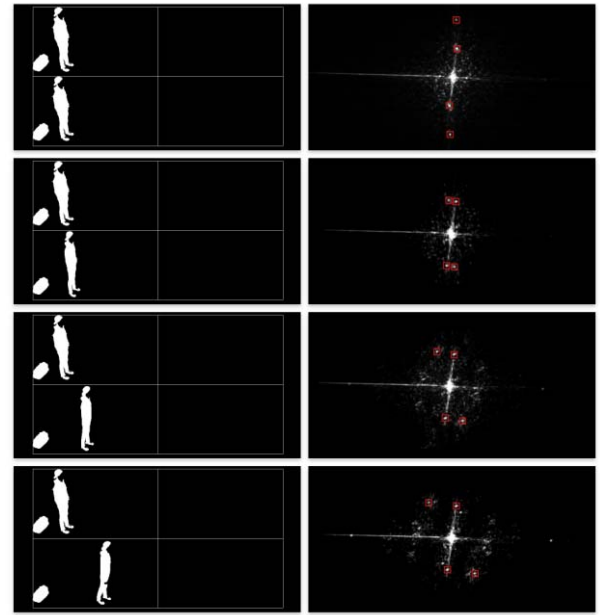


Fig. 7. Sequence of experimental frames into input of optical correlator (left) and output of optical correlator (right).

TABLE II. RESULTS OF EXPERIMENTS (1- 4)

Combination of Frames	Δx			
	Exp ₁	Exp ₂	Exp ₃	Exp ₄
1 – 1	7	7	7	7
1 – 2	7	7	8	14
1 – 3	27	25	31	34
1 – 4	49	45	52	49
1 - 5	67	70	77	71

The results of all four experiments are shown in TABLE II. The first column contains information about which two images were used in the correlation process. Other columns contain the calculated change of distance of peaks that represent luggage and silhouettes of people. Based on these values, video surveillance system evaluates movement of the monitored person. If the value of Δx has exceeded a predetermined value, the system can alert by generating a certain message type. Row 2 presents the results of the experiment (Exp₁) in Fig. 7. The

results of the other three experiments (Exp₂ to Exp₄) are given in columns 3, 4 and 5.

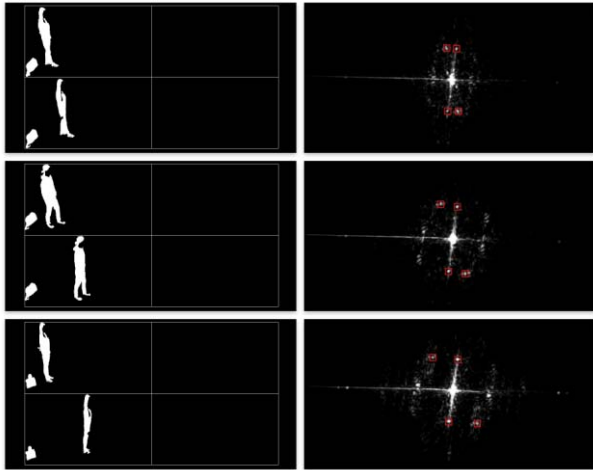


Fig. 8. Comparison of the input frames and outputs of correlator of other three experiments.

From comparison of obtained data, it can be seen that system in all four experiments recognizes the effort of people to leave luggage and move away from it. Gradually widening difference values Δx are present in each of experiments. In this way, it would be possible to obtain object location even if person tries to leave the area in a different direction. If it was the case that the correlator output will contain only two correlation peaks (correlation between luggage), the person probably completely left the monitored space [7, 8, 9, 10, 11, 12].

IV. CONCLUSION

The aim of the experiment was to use new methods in the field of digital image processing and optical correlation to design a simple video surveillance system for abandoned luggage. Its task was to detect and track the movement of objects (person, luggage) in the monitored area.

The proposed system was tested by performing the experiment, which consisted of the following elements:

- the creation of four sets of input frames that contains two persons and three kinds of luggage,
- image processing by the program created in C# - pre-processing of inputs before optical correlation,
- comparison of input frames by optical correlation,
- obtaining coordinates of objects in the monitoring area,
- evaluate changes of coordinates, and the movement of objects.

The coordinates represent the position of the correlation peaks that characterize the similarity of objects in the input scene of optical correlator. The cross-correlation of luggage between each frame was relatively high, so correlation peaks are much more evident than the correlation peaks between the silhouettes of people moving away from the luggage. The

movement of people around the scene caused a fluctuation of the value, but after a slight adjustment of camera parameters we could obtain the coordinates of these correlation peaks. Based on the changes of coordinates obtained by experiments can be evaluated and track the movement of objects in the area. After exceeding a certain critical distance, we can evaluate that person tries to leave luggage.

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REFERENCES

- [1] B. Cyganek, “Circular Road Signs Recognition with Soft Classifiers,” *Integrated Computer-Aided Engineering*, Vol. 14, No. 4, pp. 323-343, 2007.
- [2] B. V. K. V. Kumar, A. Mahalanobisbis, R. D. Juday, “Correlation Pattern Recognition,” Cambridge University Press, ISBN 9780521571036, 2005.
- [3] D. Soendoro, I. Supriana, “Traffic sign recognition with Colour-based Method, Shape-arc estimation on SVM,” in *International Conference on Electrical Engineering and Informatics (ICEEI)*, pp. 1-6, 17-19 July 2011.
- [4] C. Bahlmann, Y. Zhu, R. Visvanathan, M. Pellkofer, T. Koehler, “A system for traffic sign detection, tracking, and recognition using colour, shape, and motion information,” *Proceedings. IEEE Intelligent Vehicles Symposium*, 2005, pp. 255-260, 6-8 June 2005.
- [5] Z. Song, B. Zhao, Z. Zhu, M. Wang, E. Mao, “Research on Recognition Methods for Traffic Signs,” *FGCN '08. Second International Conference on Future Generation Communication and Networking*, 2008, pp. 387-390, 13-15 Dec. 2008.
- [6] S. Vitabile, A. Gentile, and F. Sorbello, “A neural network based automatic road sign recognizer,” presented at The 2002 Inter. Joint Conf. on Neural Networks, Honolulu, HI, USA, 2002.
- [7] P. Ambs, “Optical Computing, A 60-Year Adventure,” in *Advances in Optical Technologies Journal*, 2010.
- [8] Cambridge Correlators, <http://www.cambridgecorrelators.com>, May 2017.
- [9] Cambridge Correlators, “Fourier Optics Experimenter,” User Guide.
- [10] J. W. Goodman, *Introduction to Fourier Optics*, Colorado: Roberts & Company.
- [11] P. Ivaniga, T. Ivaniga, “10 Gbps optical line using EDFA for long distance lines”, *Przeglad Elektrotechniczny*, vol. 93, no. 3, 2017, pp. 193-196.
- [12] T. Ivaniga, P. Ivaniga, “Comparison of the optical amplifiers EDFA and SOA based on the BER and Q-factor in C-band”, In: *Advances in Optical Technologies*, doi:10.1155/2017/9053582, 2017, pp. 1-9.